Whirlpool Turbine Phase 2 Report

Executive Summary

This Phase 2 report presents the detailed CAD design and engineering analysis of the whirlpool turbine system introduced in Phase 1. The objective of this phase was to refine the conceptual design into a fully defined mechanical system and evaluate its structural performance under expected operating conditions.

The final design consists of a rotating turbine assembly composed of a central shaft and multiple turbine blades, supported within a housing that aligns and stabilizes the system. The geometry of the turbine was modified to better interact with the horizontal flow of water generated by the whirlpool, improving the transfer of fluid energy into rotational motion.

Key design decisions, including blade geometry and material selection, were guided by expected loading conditions and potential failure modes identified in Phase 1. Stainless steel was selected for the turbine blades due to its corrosion resistance and structural reliability in a water environment.

Engineering analyses were conducted to evaluate the structural integrity of critical components. These include a static stress analysis of the turbine blade and torsional analysis of the shaft. Additional considerations such as fatigue behavior, assembly, and manufacturability for 3D printing were also incorporated into the design process.

Overall, the results demonstrated that the turbine design is structurally feasible and capable of withstanding expected loads with acceptable safety margins, providing a strong foundation for prototyping in Phase 3.

Video Presentation Youtube Link:

https://youtu.be/_6kO0Aql12M

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Final Design Overview

The final turbine design consists of a rotating assembly that includes the shaft, bearings and turbine blades, enclosed within a housing designed to support and align all components. The turbine blades were redesigned to improve interaction with the horizontal water flow generated by the whirlpool. This allows for more efficient transfer of force into rotational motion, increasing the expected torque output of the system.

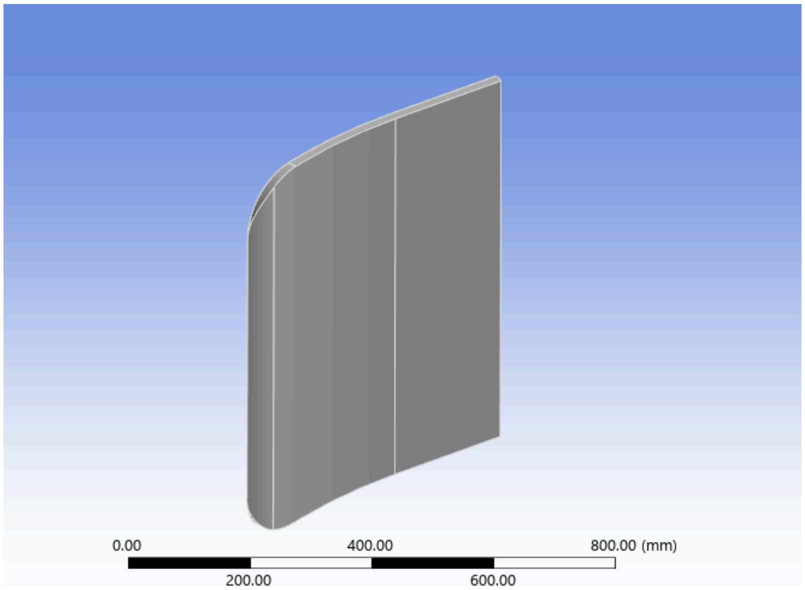

Fig 1: Stainless Steel Fan Blade

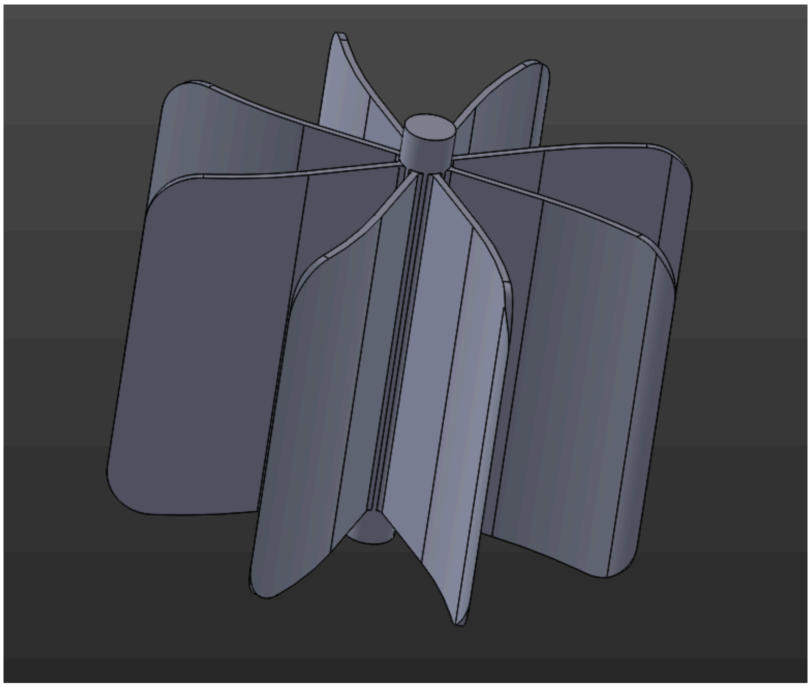

Fig 2: Shaft with Blade

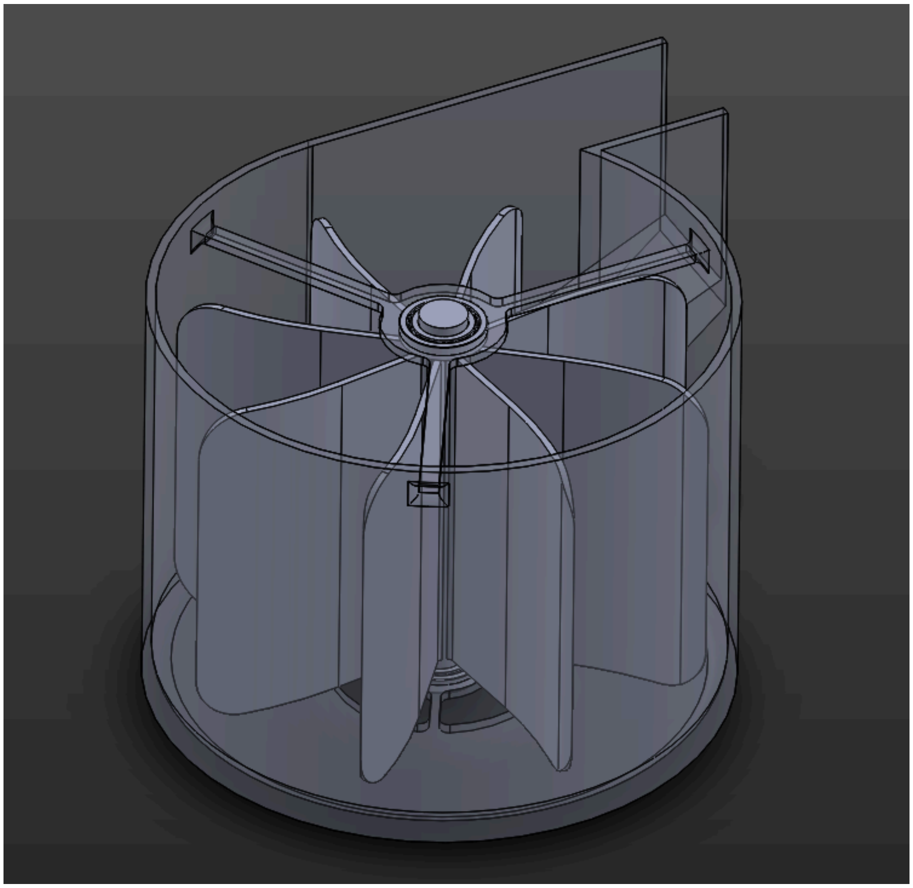

Fig 3: Complete assembly of turbine

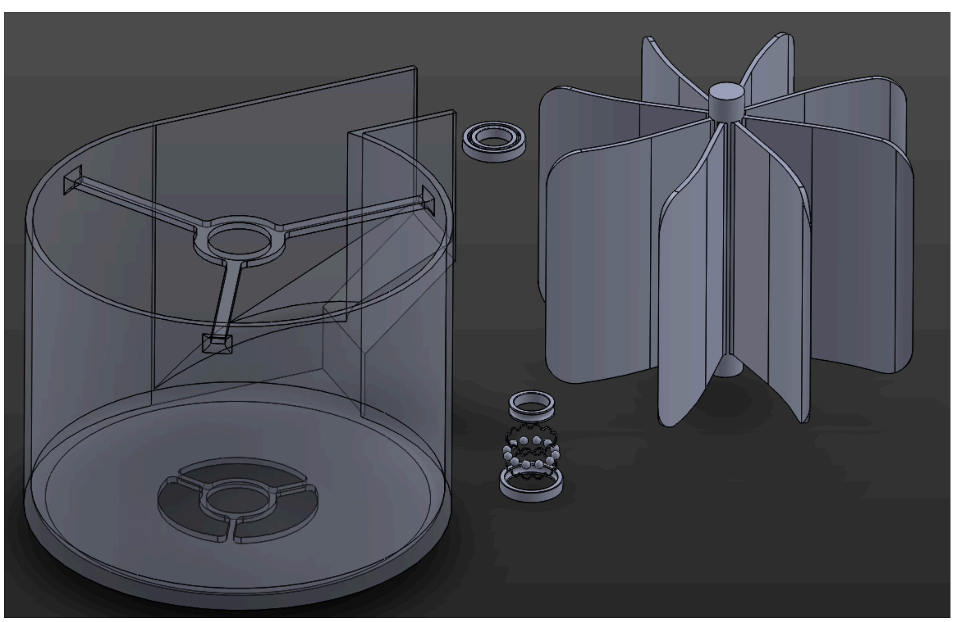

Fig 4: Complete assembly exploded view

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The CAD model represents a realistic mechanical system, with all major components positioned and constrained to reflect estimated physical behavior.

Major Design Changes from Phase 1

We decided to change the geometry of our overall turbine to better fit into the “whirlpool” assembly. The turbine blades will be longer and have their faces normal to the horizontal force of the water. Stainless steel was selected as the main material for our turbine as it has good corrosion resistance against water.

Compared to the initial conceptual design, several modifications were made:

- *Blade geometry was adjusted to increase efficiency and torque output*
- *Blade length was increased to improve water interaction*
- *Material selection was finalized (e.g., stainless steel for corrosion resistance)*
- *Structural features were added to improve strength and reduce failure risk*

These changes were made based on expected loading conditions and failure modes identified in Phase 1.)

Engineering Analyses

1. Static Stress Analyses

Turbine Blade:

A static stress analysis was performed to evaluate the structural response of the turbine blade under a worst-case loading condition. A maximum water velocity of 5 m/s was assumed.

The force acting on the blade was estimated using the drag equation:

$$F = \frac{C_d \rho_{water} V^2 A}{2}$$

Where:

- C_d = blade’s aspect ratio = 1.1825
- ρ_{water} = 1000 kg/m³
- V = velocity = 5 m/s
- A = blade area = 0.3784m

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Calculated Force:

$$F = \frac{C_d \rho_{water} V^2 A}{2} = \frac{(1.1825)(1000\frac{kg}{m^3})(5\frac{m}{s})^2(.3784m^2)}{2} = 5.593\ kN$$

The blade was modeled as a cantilever beam fixed at the shaft connection. The applied force generates bending stress, with maximum stress occurring at the blade root.

Maximum Bending Stress:

$$\sigma = \frac{Mc}{I}$$

Where:

- M = bending moment at blade root
- C = distance
- I = moment of inertia of blade cross-section

Figures 4-7 show the boundary conditions, stress distribution, deflection, and factor of safety.

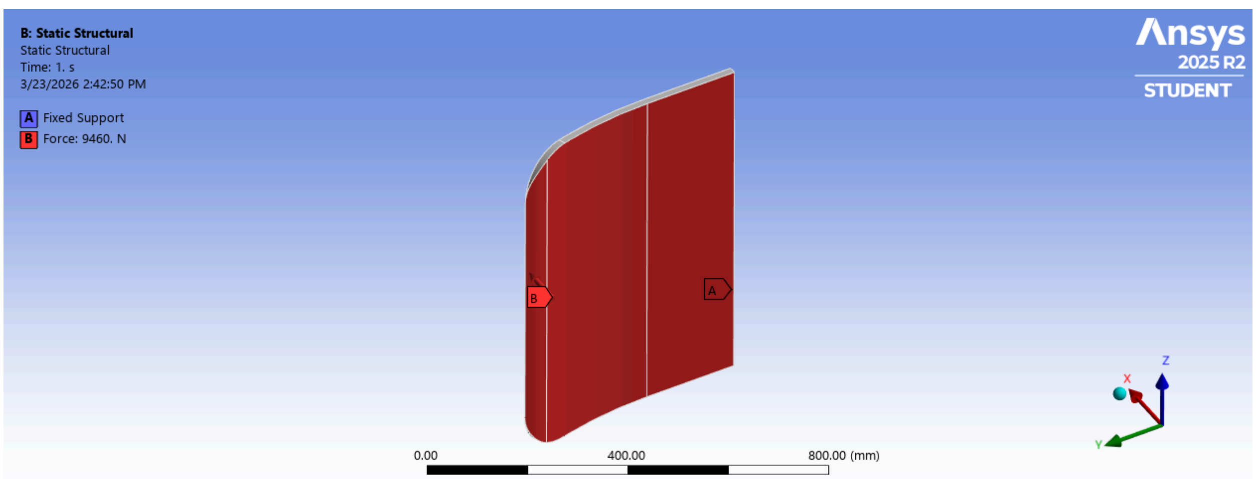

Fig 4: A fixed support is added to the unseen flat edge labeled by A. The force is applied to the entire inner edge of the blade.

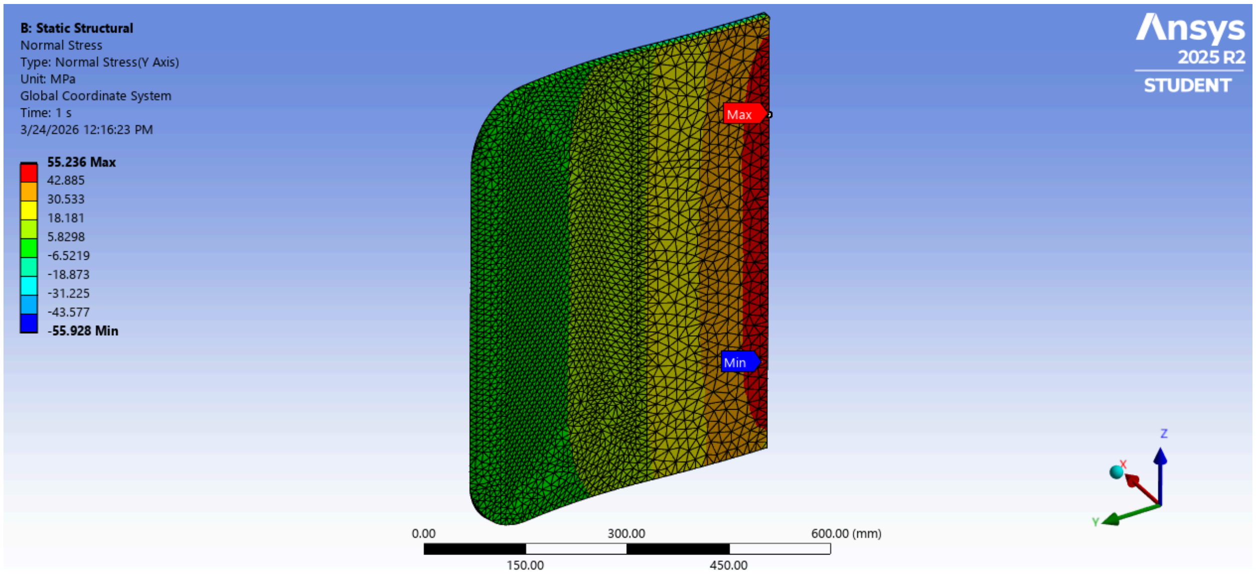

Fig 5: The max bending stress occurs where the blade would be fixed to the shaft as expected.

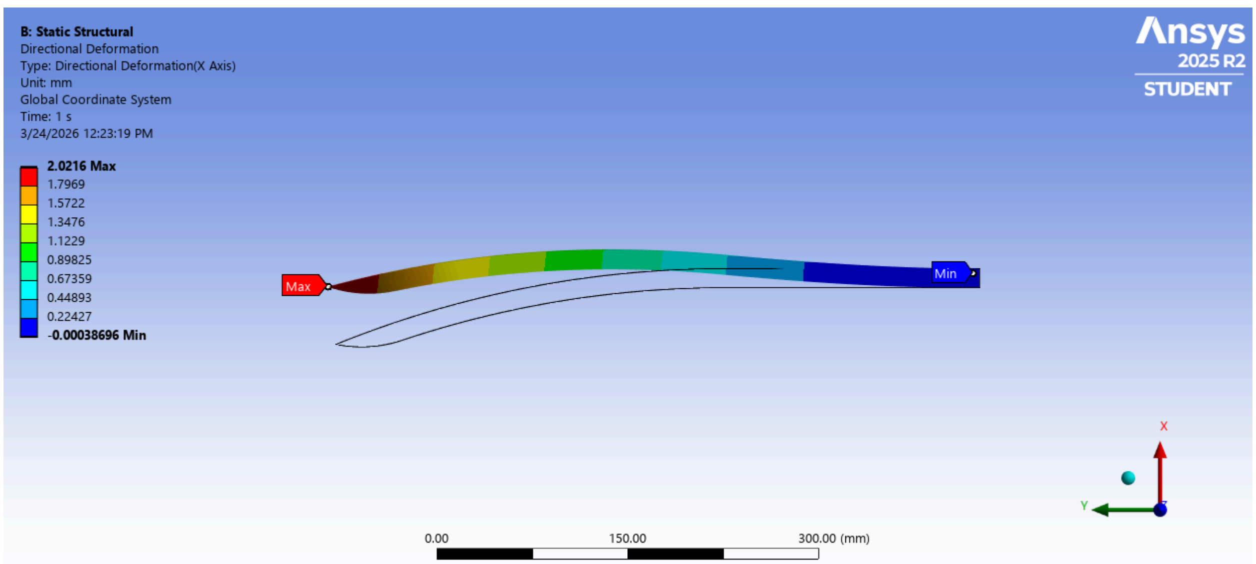

Fig 6: The deflection, as exaggerated in the above image, is very small.

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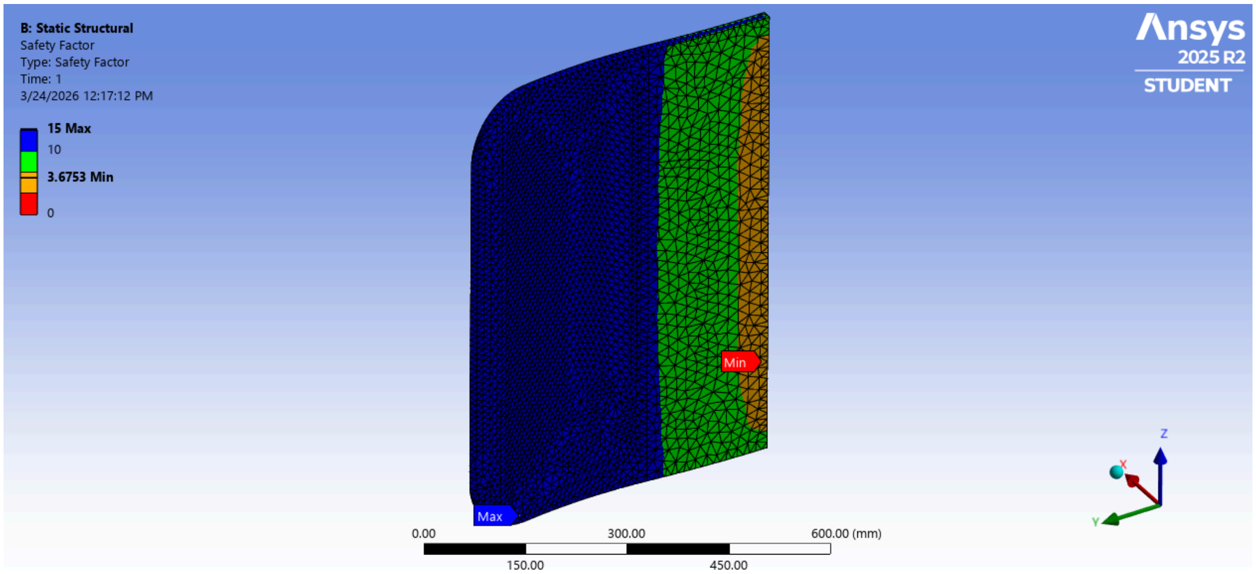

Fig 7: The factor of safety (about 3.7) is optimal.

Results:

- Maximum stress: 55.236 MPa
- Minimum Factor of safety: 3.6753
- Maximum Deflection: 2.0216 mm

The factor of safety indicates that the blade is structurally adequate under expected loading. Also, the max stress and deflection indicates the design of the turbine is structurally adequate as well.

Bearing / Interface:

The two bearings will experience a force pointing downward on the inner face of the bearing, due to the weight of the runner connected to the bearing.

Reaction force at shaft support:

- $F_{runner} = m_{runner} g$ = 3924 N
- $\omega = \frac{v}{r} = \frac{(5\frac{m}{s})}{(0.295\ m)}$ ~ 17 rad/s

Figures 8 - 11 show the boundary conditions, stress distribution, deflection, and factor of safety.

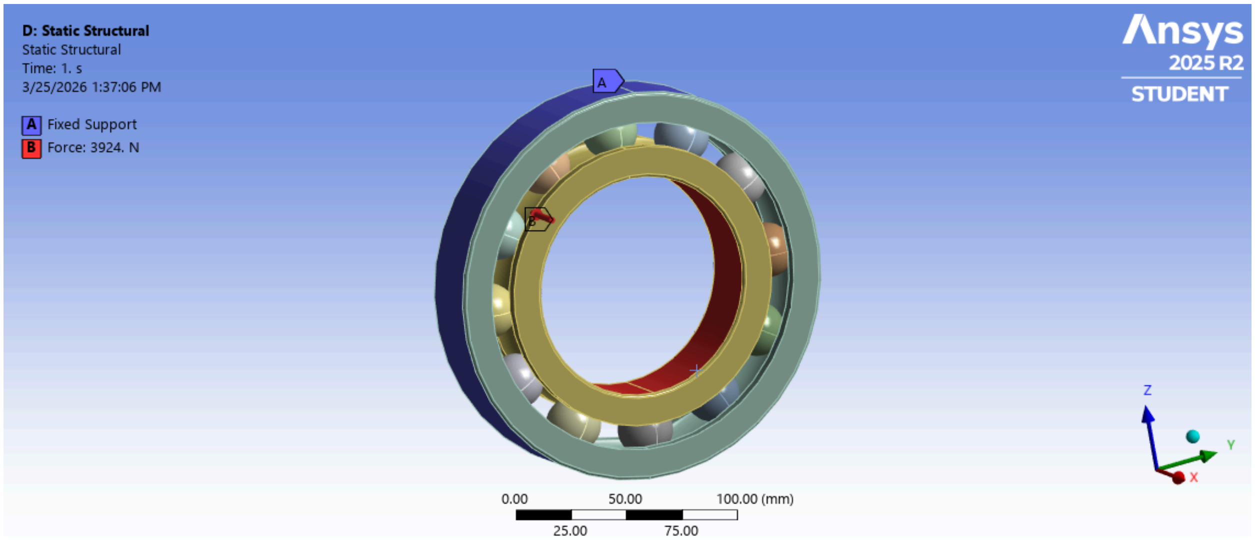

Fig 8: Shows the bearing fixed at A and the force is applied to the inner surface of the bearing pointing in the opposite direction of support for the shaft.

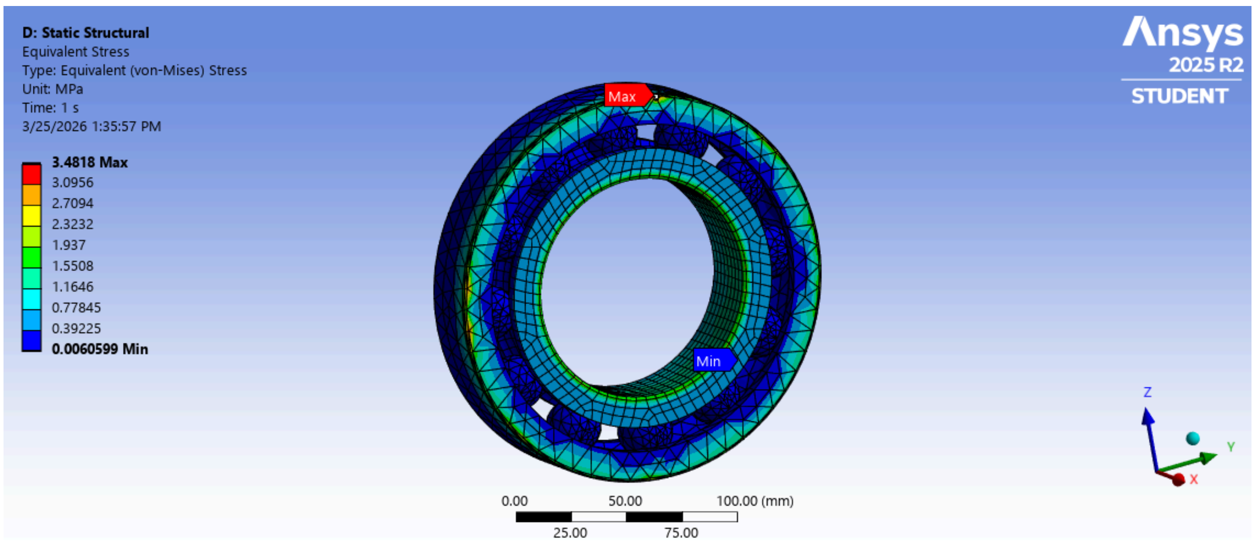

Fig 9: The maximum stress occurs at the outermost edge of the bearing.

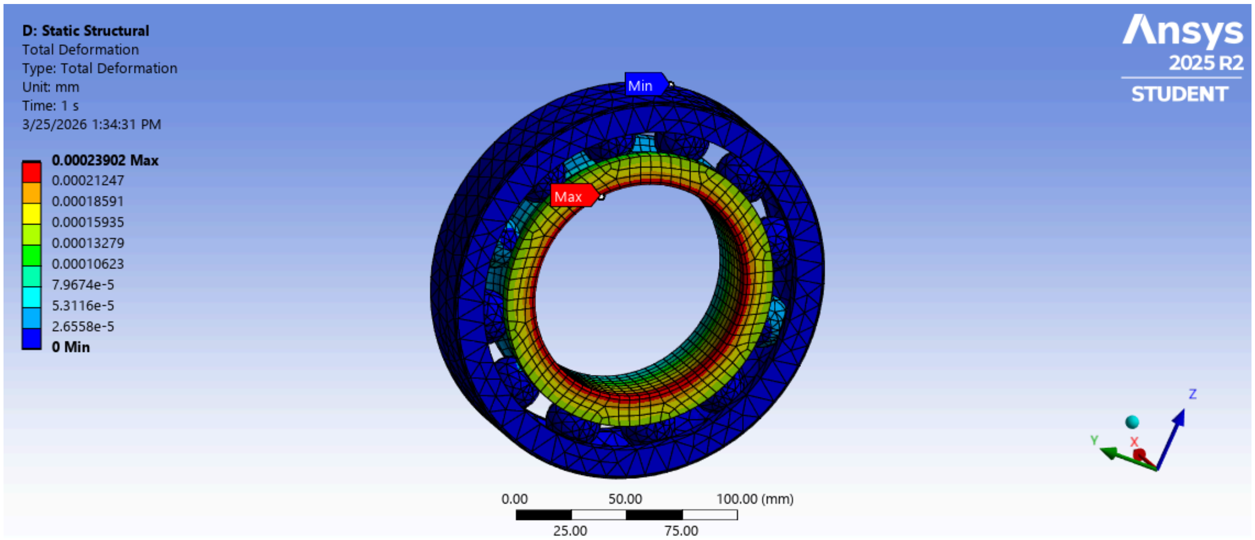

Fig 10: The deformation due to the weight of the runner on the inner face of the bearing is very small.

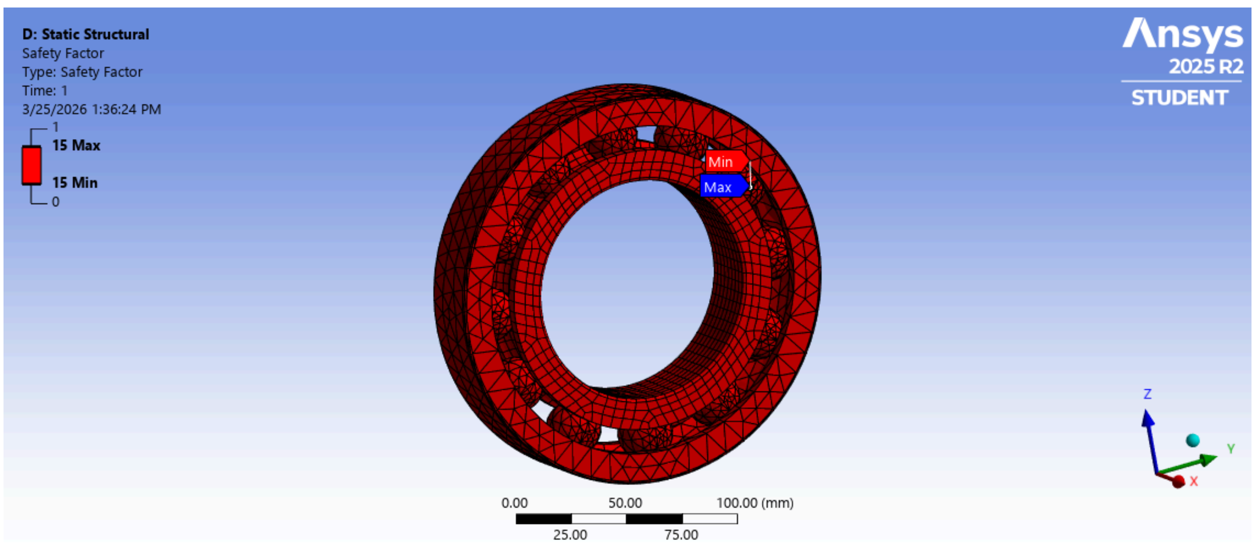

Fig 11: The bearing has a really high factor of safety.

2. Shaft Torsion Stress Analysis

The torque applied to the shaft is generated by the force acting on the turbine blades. The maximum torque applied to the shaft occurs under peak loading conditions occurring when the water initially starts flowing. This torque is estimated using:

$$T = F \cdot r$$

Where:

- F = force exerted by water on one initial blade (N)
- r = turbine radius + shaft (m)

Using the previously calculated force:

$F = 5593\ N$

$r = 0.295\ m$

$T\ =\ 5593N\cdot(.05\ +.245)$

Torque: $1650\ N\ *\ m$

Assumptions:

- The shaft is solid and circular.
- Torque is constant along the shaft.

Figures 12 - 15 show the boundary conditions, twist deformation, shear stress distribution and factor of safety.

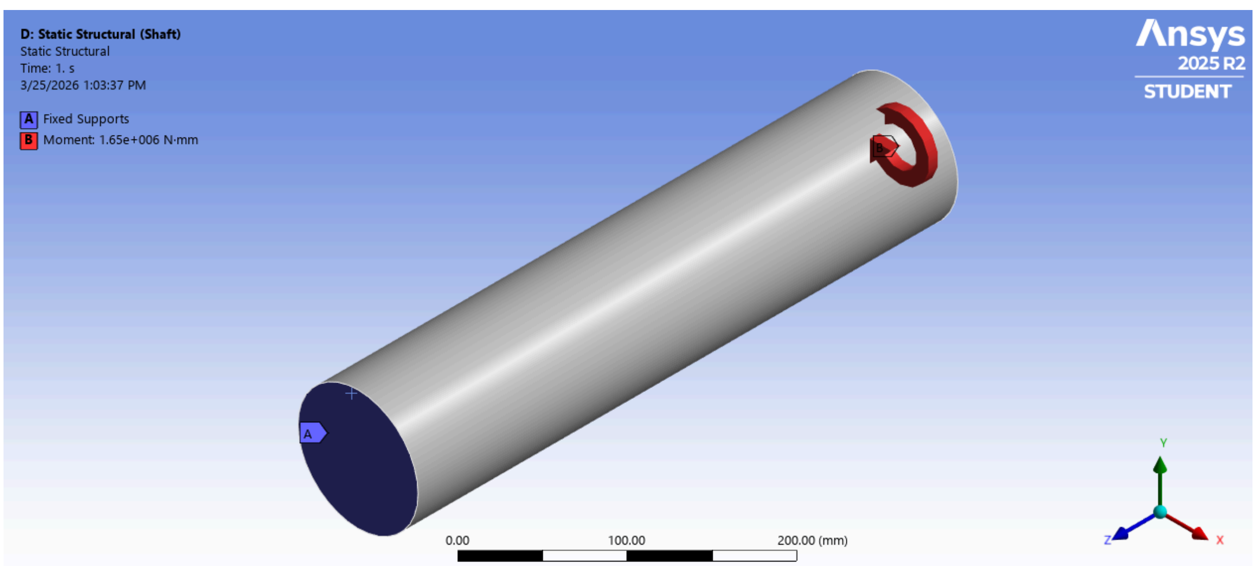

Fig 12: Shows shaft fixed at A and the moment is applied to the middle of the shaft at B using a symmetry condition ("Cutting" off half the shaft for computational efficiency).

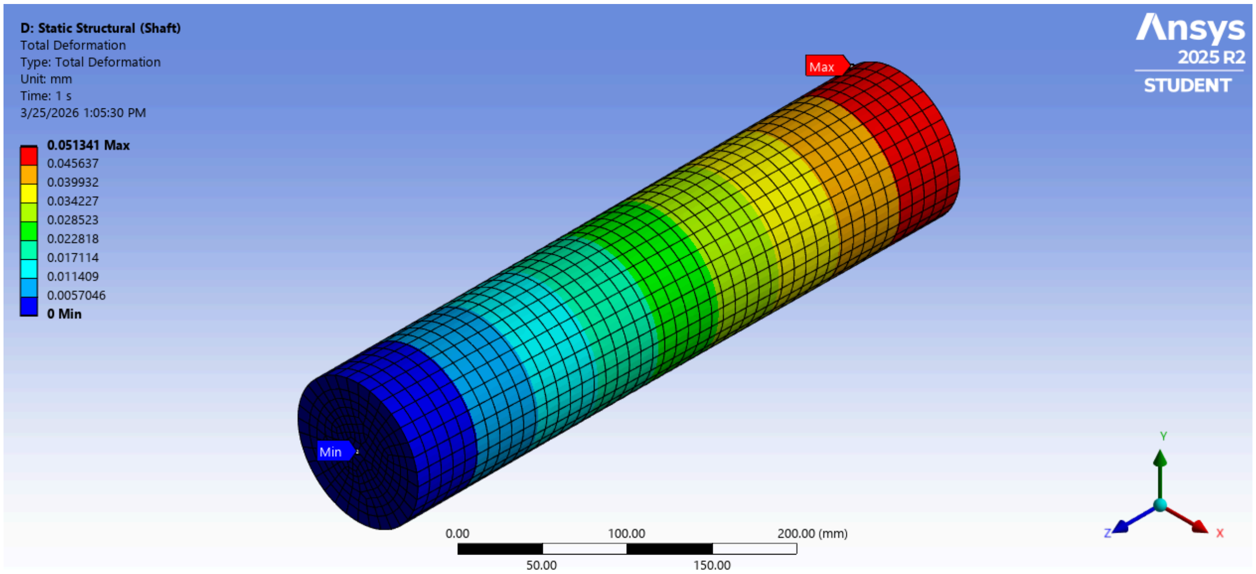

Fig 13: The angle of twist is small since the overall deformation is only a few centimeters.

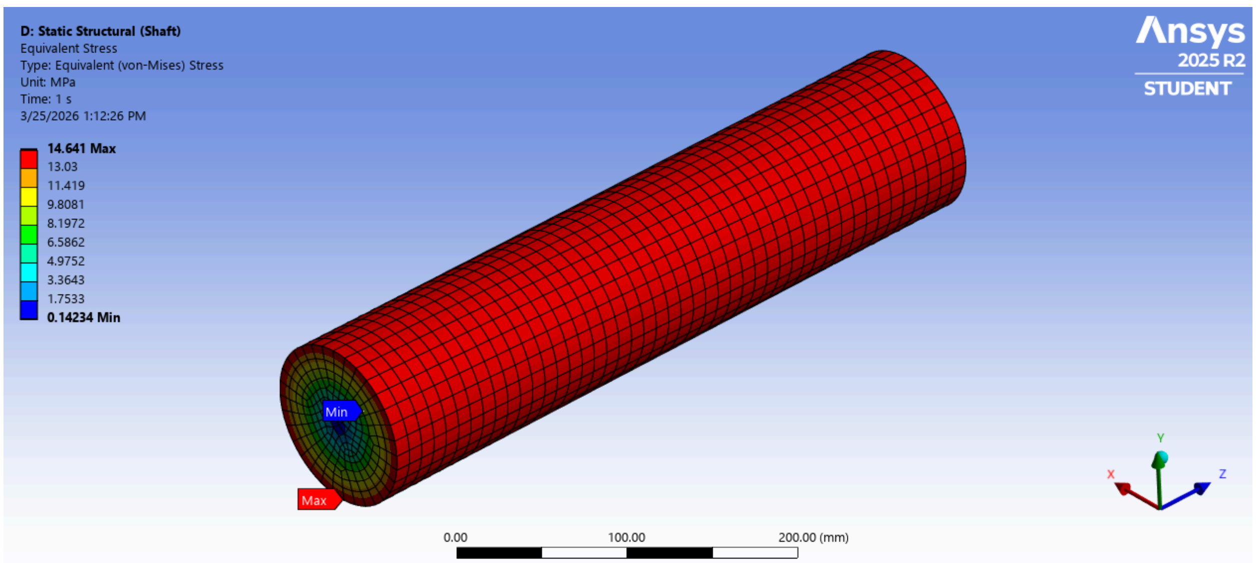

Fig 14: The maximum stress occurs at the outermost surface of the shaft as expected.

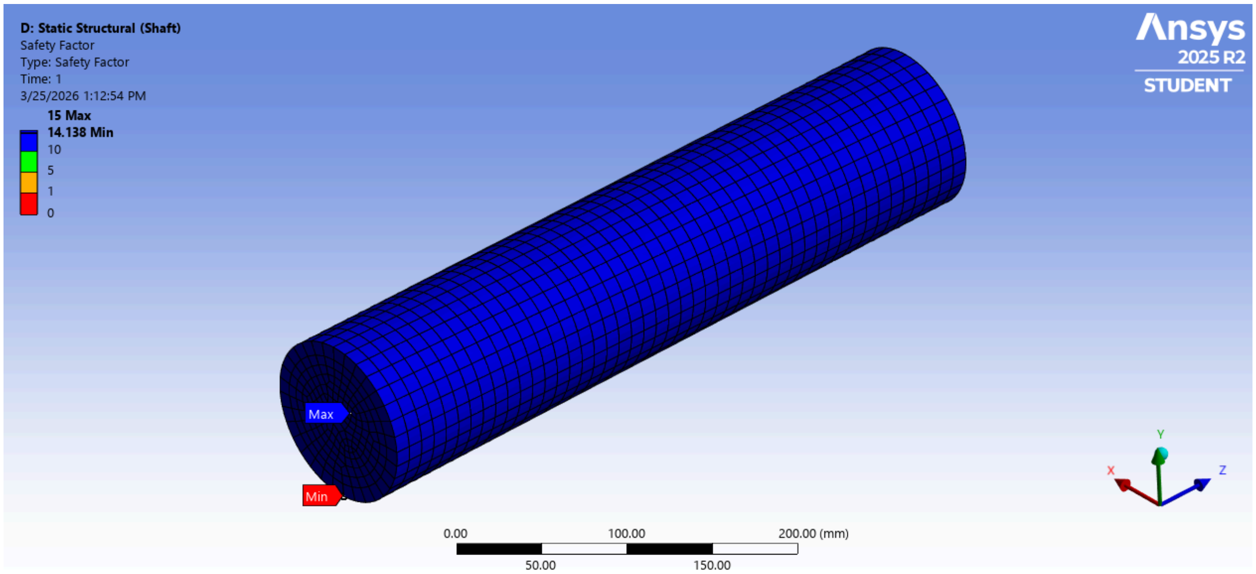

Fig 15: The shaft has a really high factor of safety.

Results:

This analysis shows that shaft diameter is critical in preventing torsional failure. The calculated max stress is high but, the factor of safety is still really high. This is probably due to the fact that our shaft is pretty thick and the simulation is only accounting for a maximum force from a small river flow.

3. Fatigue Analysis (Turbine Blade)

The turbine is expected to experience repeated loading every time a blade is exposed to the water force, making fatigue an important design consideration.

Loading conditions:

- The loading fluctuates between zero and maximum stress.
- Mean stress from steady rotational operation.

A simplified fatigue analysis on a single blade was performed to ensure safe operation over time. The turbine blade experiences repeated loading and since the stress levels are low relative to material limits it is expected that fatigue failure is unlikely.

Design considerations:

- Smooth surface finishes to reduce crack initiation.

- Fillets at geometric transitions to reduce stress concentration.
- Uniform shaft geometry improves fatigue life.

A more detailed fatigue analysis will be conducted in Phase 3 once more accurate loading data is available.

Figures 16 - 18 show the Fatigue Life, Damage and Safety Factor of the blade. (The same boundary conditions were used as in Figure 4 above).

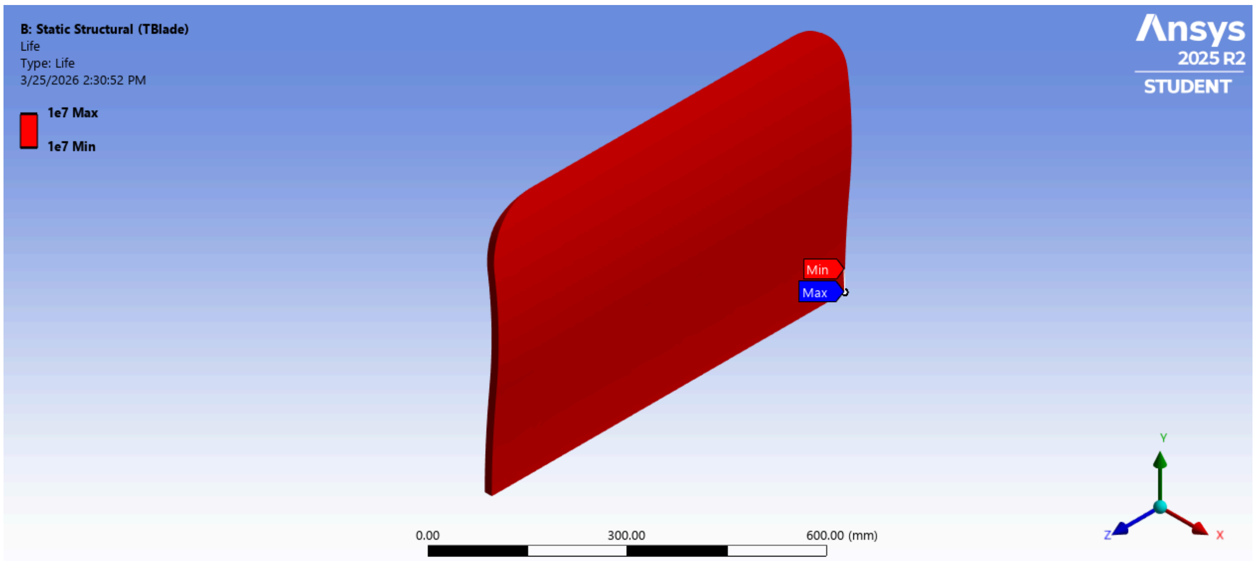

Fig 16: "Infinite" life achieved as desired

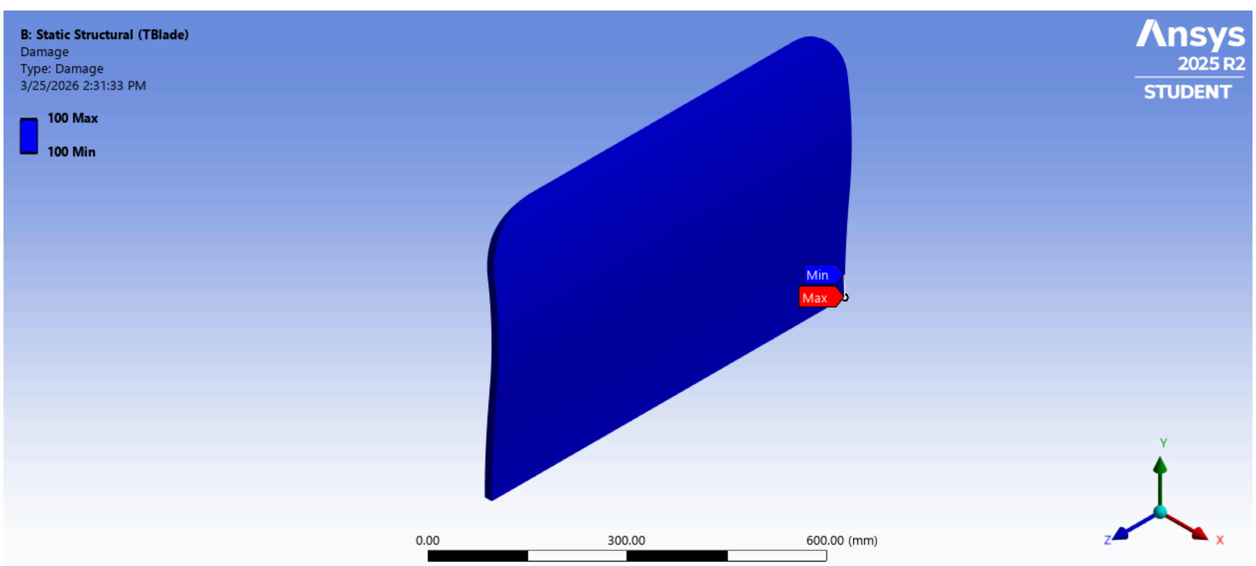

Fig 17: No damage achieved as desired

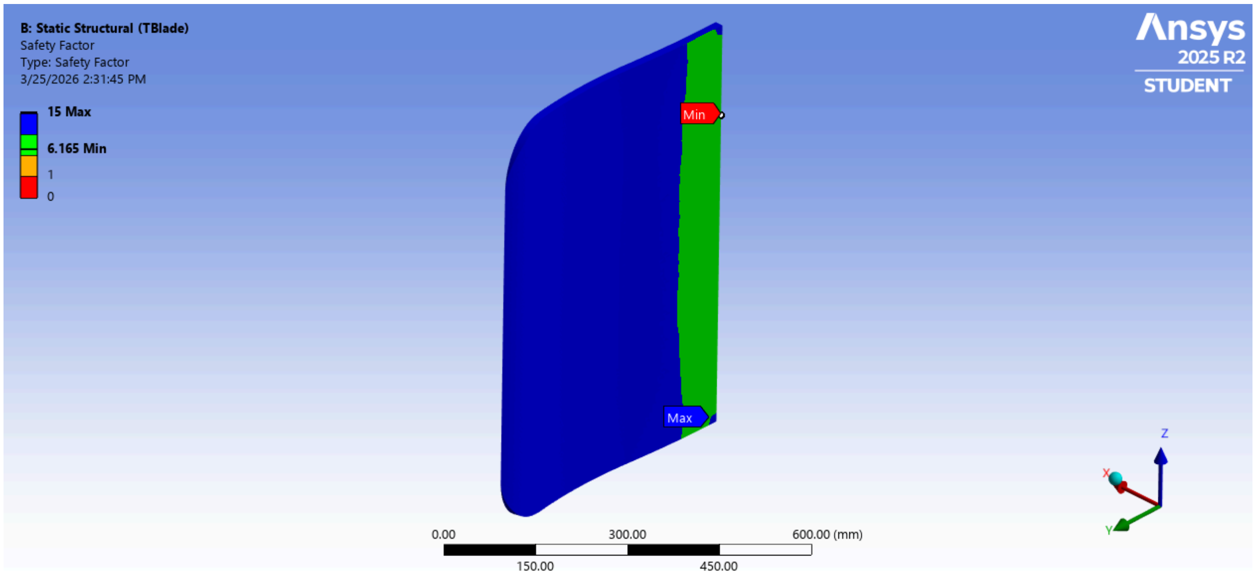

Fig 18: The blade's fatigue FoS is optimal.

4. Summary

The following table summarizes the structural performance of key components:

Component	Max Stress	Factor of Safety (Static)	Factor of Safety (Fatigue)	Status
Turbine Blade	$55\ MPa$	3.6753	6.165	Safe
Shaft	$14.6\ Mpa$	14.1	N.A. (Blade will fail before shaft)	Safe
Bearings	$3.48\ MPa$	15	N.A. (Blade will fail before bearings)	Safe

Discussion:

The shaft and bearings demonstrate adequate strength under static loading conditions while the turbine blades demonstrate adequate strength and durability under static and repeated loading conditions. This is optimal as the turbine blades are the weak point in the system.

Design for Assembly

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The turbine was designed with a modular approach to simplify assembly:

- Components are modular and can be assembled individually
- Shaft and blades are connected using press-fit or adhesive methods
- Alignment features are included to ensure proper positioning
- The housing supports and stabilizes the rotating components

This approach improves manufacturability and ease of maintenance.

Design for 3D Printing

The CAD model should be adjusted to meet 3D printing constraints:

- Parts are scaled to fit printer build volume
- Components are separated into printable sections
- Clearances were added for proper fit
- Overhangs were minimized to reduce supports

Additional considerations:

- Blade thickness increased for strength
- Print orientation was selected to improve strength

Updated Risks and Weaknesses

Based on this design phase, the following risks remain:

- Potential misalignment during assembly
- Fatigue failure over time
- Friction losses in rotating components
- Scaling inaccuracies between model and real system

Conclusion

This Phase 2 report demonstrates the progression from a conceptual design to a fully defined mechanical system. Through CAD modeling and engineering analysis, the turbine design has been validated for structural performance and manufacturability.

The results show that the design is capable of withstanding expected loads with acceptable safety margins. Future work in Phase 3 will focus on prototyping, testing and evaluating the turbine’s motion under simplified real-world conditions.

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References/Citations

Budynas, Richard G, et al. *Shigley’s Mechanical Engineering Design*. New York, Mcgraw-Hill, 2016.

Çengel, Yunus A, and John M Cimbala. *Fluid Mechanics : Fundamentals and Applications*. 4th ed., Singapore, Mcgraw-Hill Education, 2020.

Hibbeler, R. C. Mechanics of Materials. 11th ed., Pearson, 2017